

CYTOLOGY

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Proteasomes:

- They are small organelles composed of protein complexes that are responsible for **proteolysis of malformed proteins.**
- Proteins as those of the metabolic regulation have to be degraded to ensure that the metabolic response to a signal stimulus is not prolonged.
- Also **denaturated, damaged or malformed** proteins have to be eliminated.
- the process of proteolysis is carefully controlled by the cell.

B. Non membranous organelles

1- Ribosomes

2- Centrioles

Ribosome

- It is a small electron-dense particle, about 20 x 30 nm in size found in the cytoplasm **free** or **fixed** to endoplasmic reticulum forming rER.
- It has **2 subunits** (large subunit and small subunit) and formed of ribosomal RNA and ribosomal proteins. The 2 subunits are present in the **cytoplasm** individually and do not form a ribosome until protein synthesis begins.

Ribosome

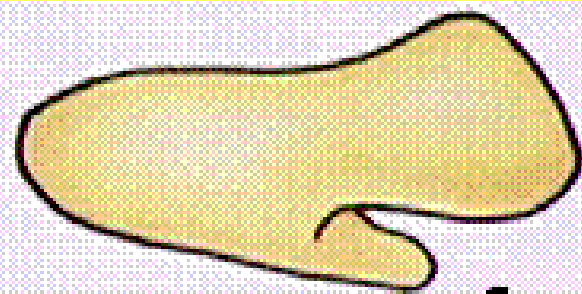
- The **rRNA** is synthesized **within the nucleolus** while the **proteins** are synthesized in the **cytoplasm** and then enter the nucleus to be associated with the rRNA.
- Subunits then leave the nucleus to enter the cytoplasm.
- Ribosomes can't be seen by LM but they give **basophilia** if they are abundant in the cytoplasm.

- The **free** ribosomes produce proteins to be **used inside the cell** eg. **erythroblasts** which produce hemoglobin, meanwhile the **fixed** ribosomes produce proteins to be **secreted from the cell** eg. **salivary enzymes** or stored in the cell eg. enzymes of lysosomes.

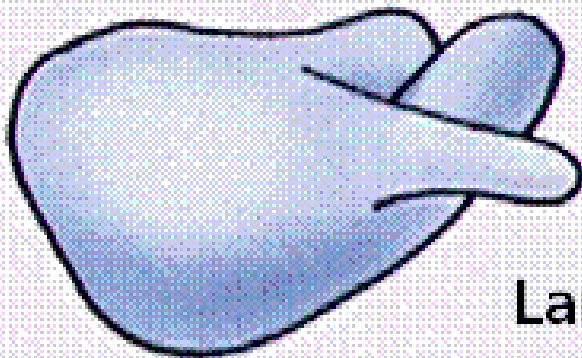
N.B.

There are 3 types of RNA (ribonucleic acid):

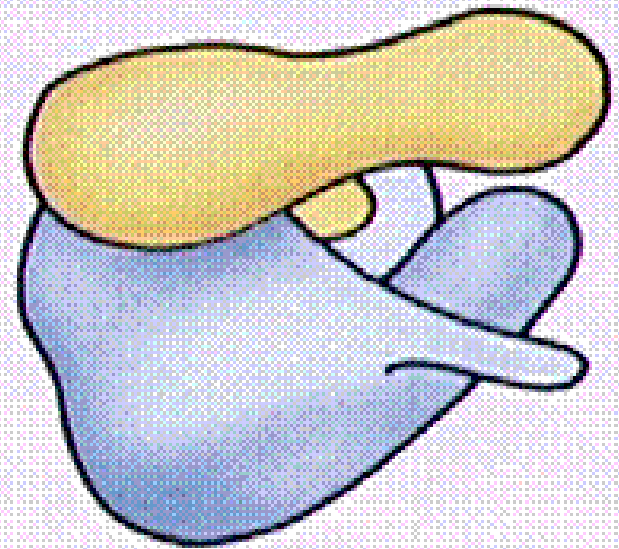
- 1- rRNA (ribosomal RNA)**
- 2- mRNA (messenger RNA)**
- 3- tRNA (transfer RNA)**

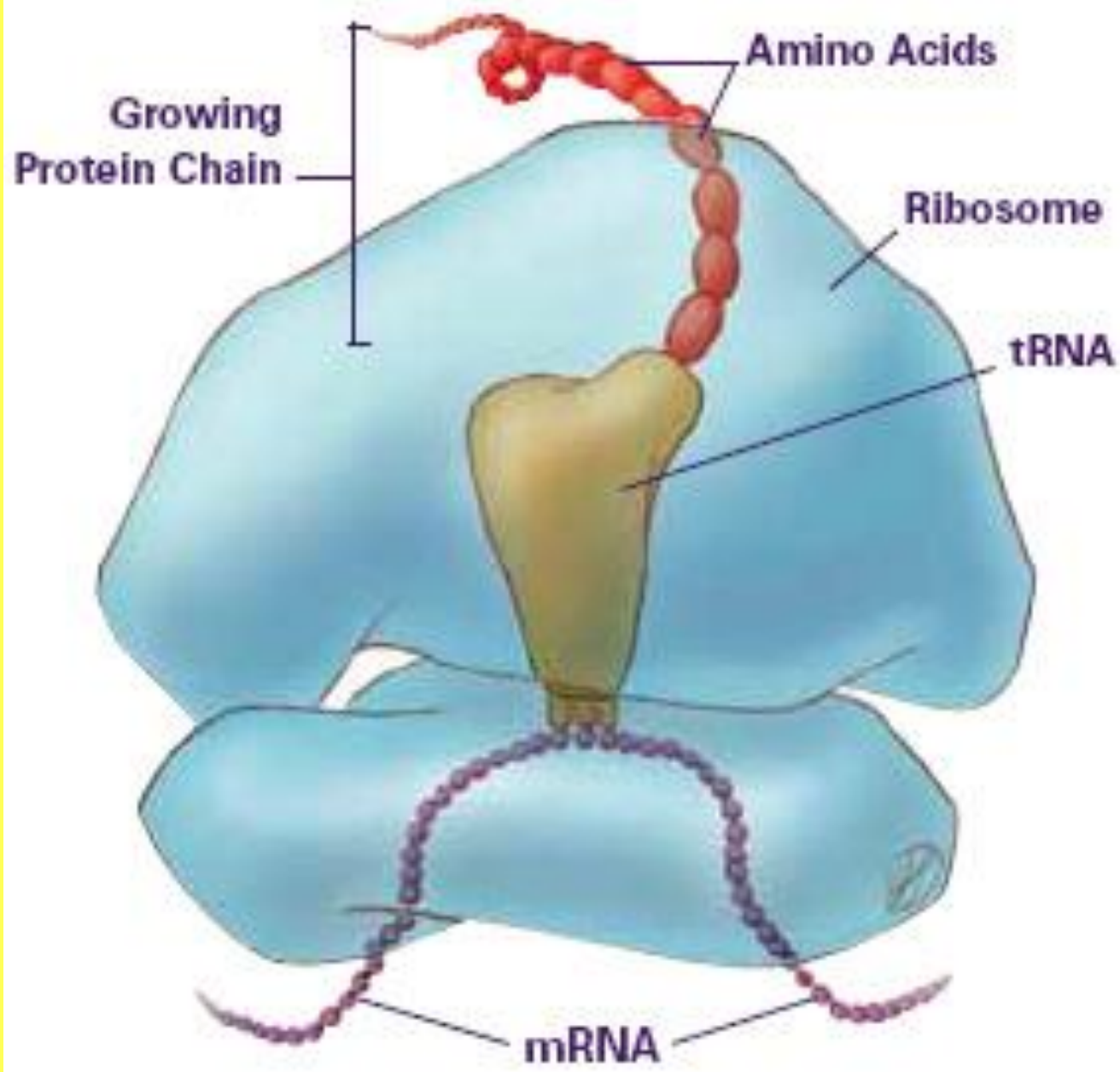


Small subunit



Large subunit





Centrosome and centrioles

- The centrosome (**microtubule organizing center**) are composed of **2 centrioles** arranged **perpendicular** to each other and the surrounding material.
- By LM the centrosome appears as 2 dark dots.

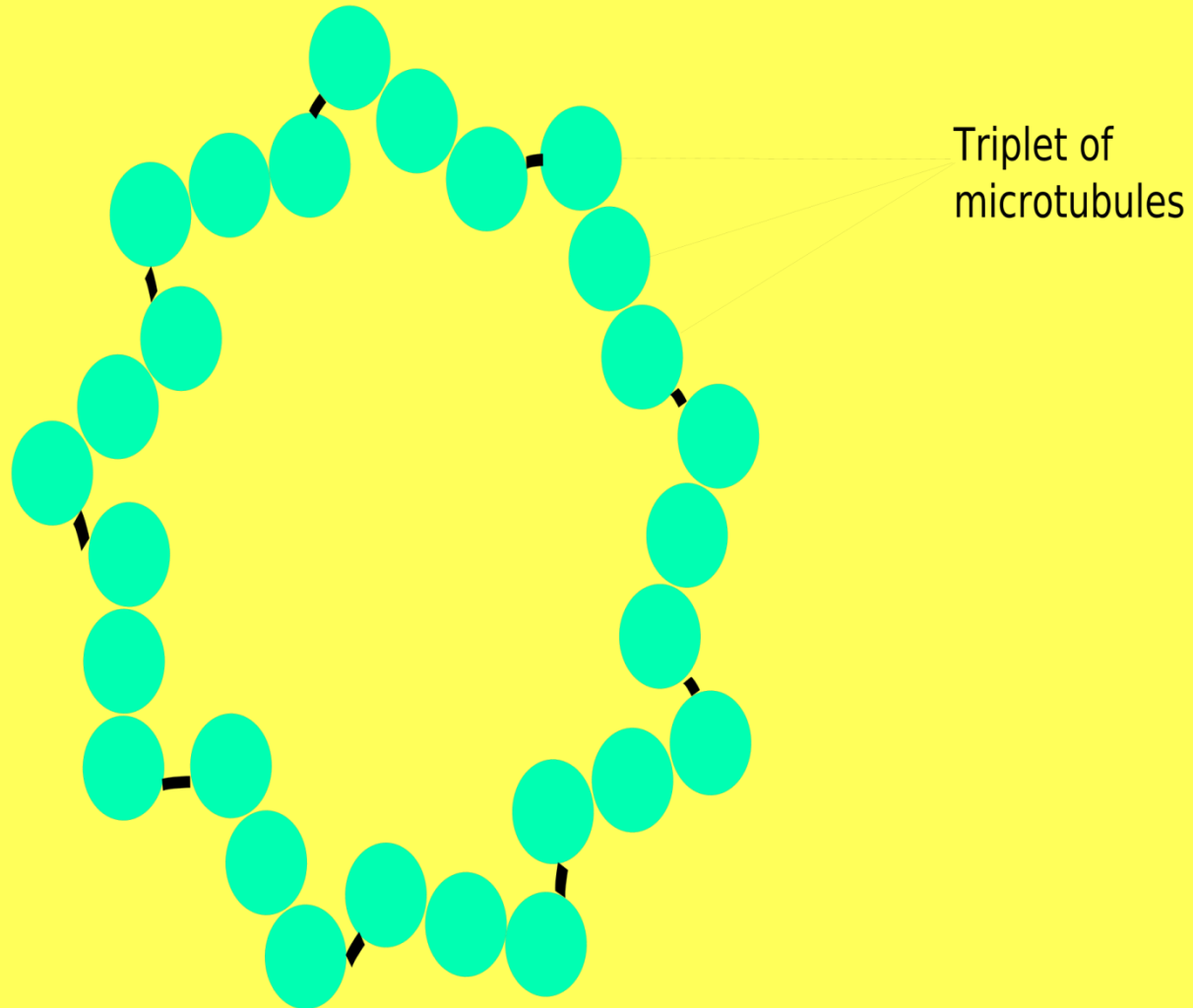
✓ Centrioles

- Each centriole is small cylindrical structure $0.2\ \mu\text{m}$ in diameter and $0.5\ \mu\text{m}$ in length , closed at one end while the other end is opened.
- - The wall of the cylinder is consisted of **nine triplets** of parallel microtubules. The inner one is circular and the other 2 are incomplete (c-shaped).

- Each of the inner microtubules is connected to the outer most one of the adjacent triplet by **fine filament**, thus **the 9 triplets form a cylinder**.

Functions

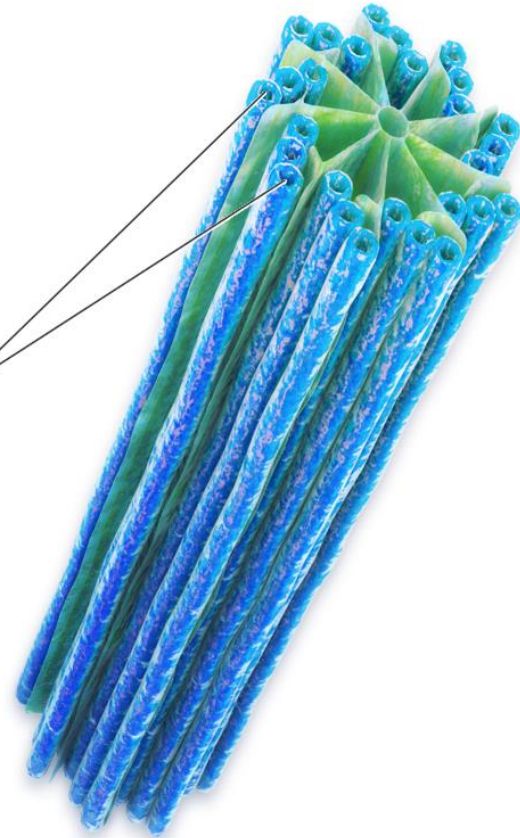
- 1- They form the centrosome.
- 2- They form the mitotic spindle in cell division.
- 3- They form the basal bodies of the cilia and flagella.



Cross section of centriole - Schema



Microtubules



Centrioles

Cytoskeleton

- The cytoplasmic cytoskeleton is a complex network of **microtubules**, **microfilaments** and **intermediate filaments**.
- They are structural proteins having a functions in:
 - Maintenance of the cell shape.
 - Movement of organelles and intracytoplasmic vesicles.
 - Movement of the entire cell.

Microtubules

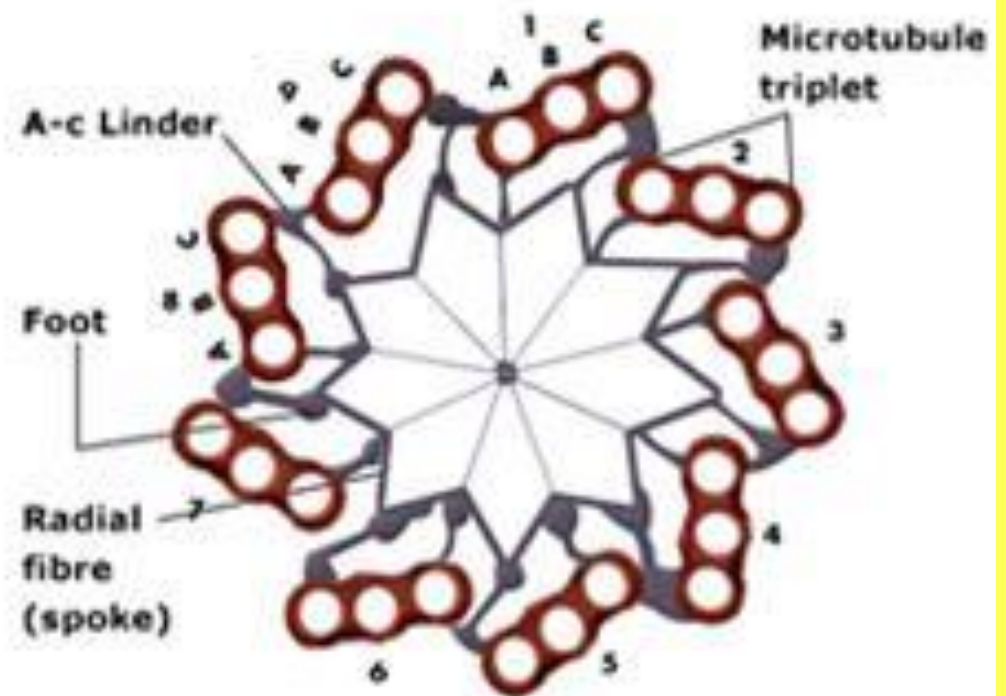
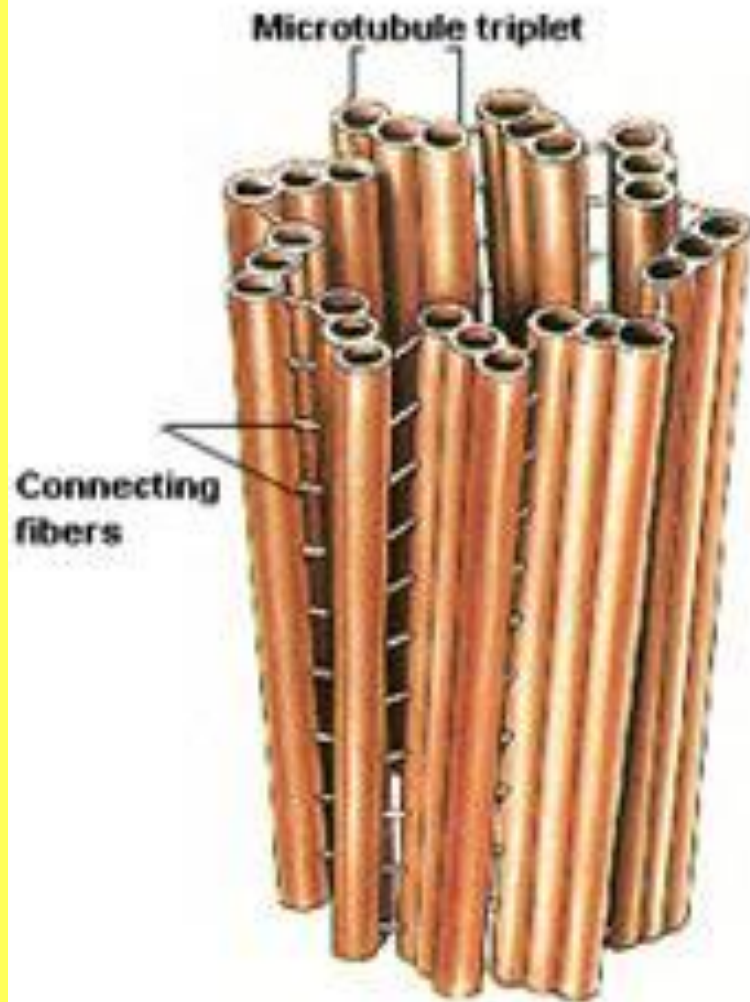
- They are long, straight, rigid, hollow-like cylindrical structure 25 nm in diameter with variable lengths may reach several micrometers.
- They are polarized having a rapidly growing **plus end** and a **minus end** which must be stabilized or it will depolymerize, thus shorting the microtubule.
- The minus end is stabilized by being **embedded in the centrosome**.

Microtubules

- They are dynamic by changing their length by growth and shorting. Both processes occur at the plus end.
- Each microtubule consists of 13 **protofilaments** composed of heterodimers of polypeptide α and β tubulin subunits.
- Polymerization of tubulin to form microtubule occurs by the **microtubule organizing centers** (a variety of structures include basal bodies, centrioles and the centromeres of the chromosomes).

Microtubules

- **The stability of the microtubules is variable, ex. In the cilia they are stable whereas those of the mitotic spindle have a short duration (20 minutes).**
- **Tubulin polymerization is under the control of the concentration of Ca^{++} and the microtubule associated protein (MAPs).**



Microtubules

Functions

1. They provide rigidity and maintain cell shape.
2. They regulate intracellular movement of organelles and vesicles.
3. Establish intracellular compartments.
4. Segregation of daughter chromosome at mitosis.
5. They generate sliding forces between the adjacent doublet microtubules of cilia by which the cilia generate ciliary motility.

N.B.

- Microtubules provide the basis for several complex cytoplasmic components including, centrioles, basal bodies, cilia and flagella.

Microfilaments (Thin filaments)

- They are protein **Actin** in muscles they are composed of two chains globular subunits **G-actin** coiled around each other to form a filamentous protein **F-actin**.
- The filament have 5-7 nm thick and possess a faster-growing **plus end** and a slower-growing **minus end**.
- There are 3 classes of actin: **α -actin** of muscle and **β -actin** and **γ -actin** in non-muscle cells.

Microfilaments (Thin filaments)

- **Myosin** is the most binding-protein that interact with actin to bring about intracellular or cellular movement.
- **Actin** is present in all cells.
- **Actin can be organized within the cells in many forms:**
 1. In **skeletal muscle** they form thin filaments .
 2. In **most cells** they form a thin sheath just beneath the cell membrane called the **cell cortex**. These filaments are associated with endocytosis, exocytosis and cell migration.

Microfilaments (Thin filaments)

3. Microfilaments associated with the **cytoplasmic organelles**.
4. In most cell they are found **scattered** every where in the **cytoplasm**.

Intermediate filaments

- They are **proteins** having a 10-12 nm diameter.
- They provide:
 1. Support for the cell.
 2. They form three-dimensional framework for the cells.
 3. Anchor the nucleus in place .
 4. Connect between the cell membrane and the cytoskeleton.

Intermediate filaments

- They include:

1. Keratins

- They are found in most epithelia.

2- Vimentin

- Found in cells of mesenchymal origin.

3- Desmin (skeleton)

- Found in smooth muscle and Z disks.

4- Glial filaments

- Found in astrocytes (one of the supporting cells of the nervous tissue)

5- Neurofilaments

- Found in nerve cells (neurons)

Cytoplasmic inclusions

- They are materials found **temporary** in the cytoplasm and not considered a part of the cell eg. **stored foods** and **pigments**.
- **Stored foods**
- **Carbohydrate** and **fat** are stored in certain cells as inclusions.
- **Carbohydrate** is stored in the form of **glycogen** (free deposits) in the cytoplasm of hepatocytes and muscle cells.
- **Fat** is stored by fat cells of adipose tissue and in the form of free cytoplasmic droplets (**without limiting membrane**) in the hepatocytes.

Cytoplasmic inclusions

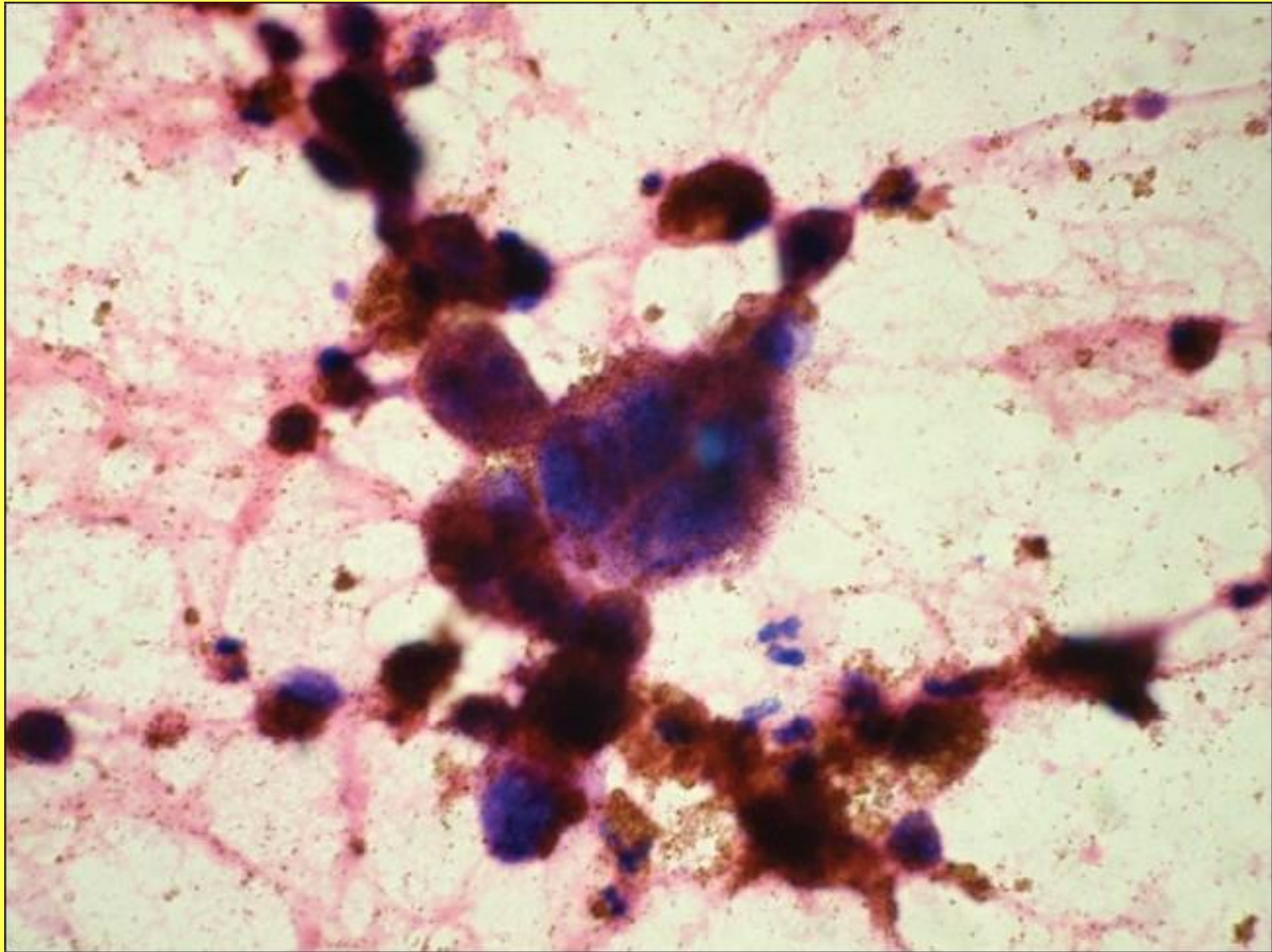
- Pigments
- There are two groups, **exogenous** and **endogenous** pigments.

Exogenous pigments:

- Those are the pigments which are formed **outside** the body and taken into it.
eg. **carotene** pigments, carbon particles (**dusts**) and **tattoo**.

Endogenous pigments:

- The pigments which are made up **inside** the body.
eg. **melanin**, **lipofuscin** and **haemosiderin** pigment.



2- Nucleus

- Most of the cells contain a nucleus varies in appearance from cell to another, but in general has a **nuclear envelope**, **chromatin**, **nucleoplasm** and one or more **nucleoli**.
- Nuclei display wide variations in: shape, size, number per cell, location and the chromatin pattern.
- The nucleus contains **DNA** (about 20%) associated with nucleoprotein (histones and non-histones) in addition to some **RNA**.

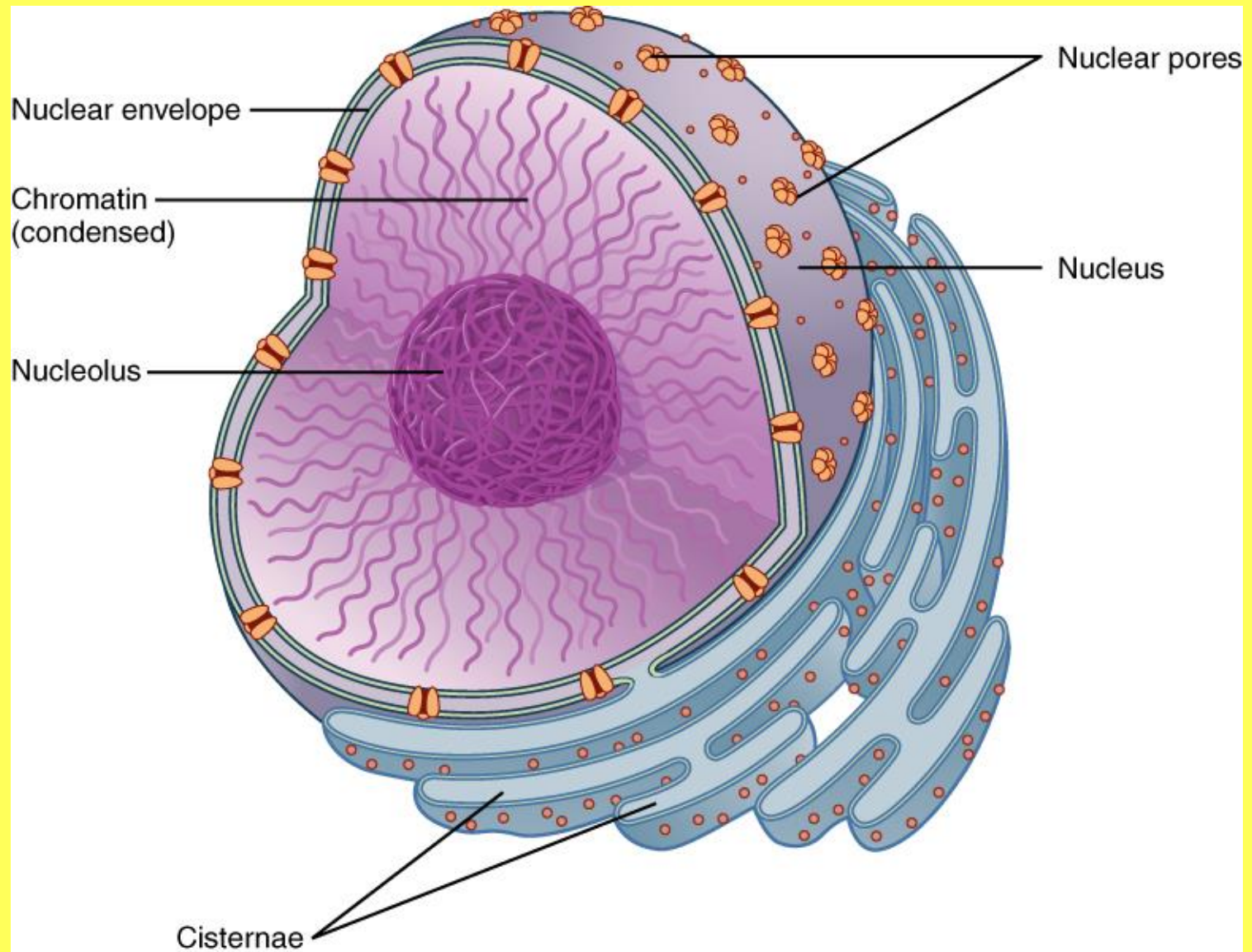
2- Nucleus

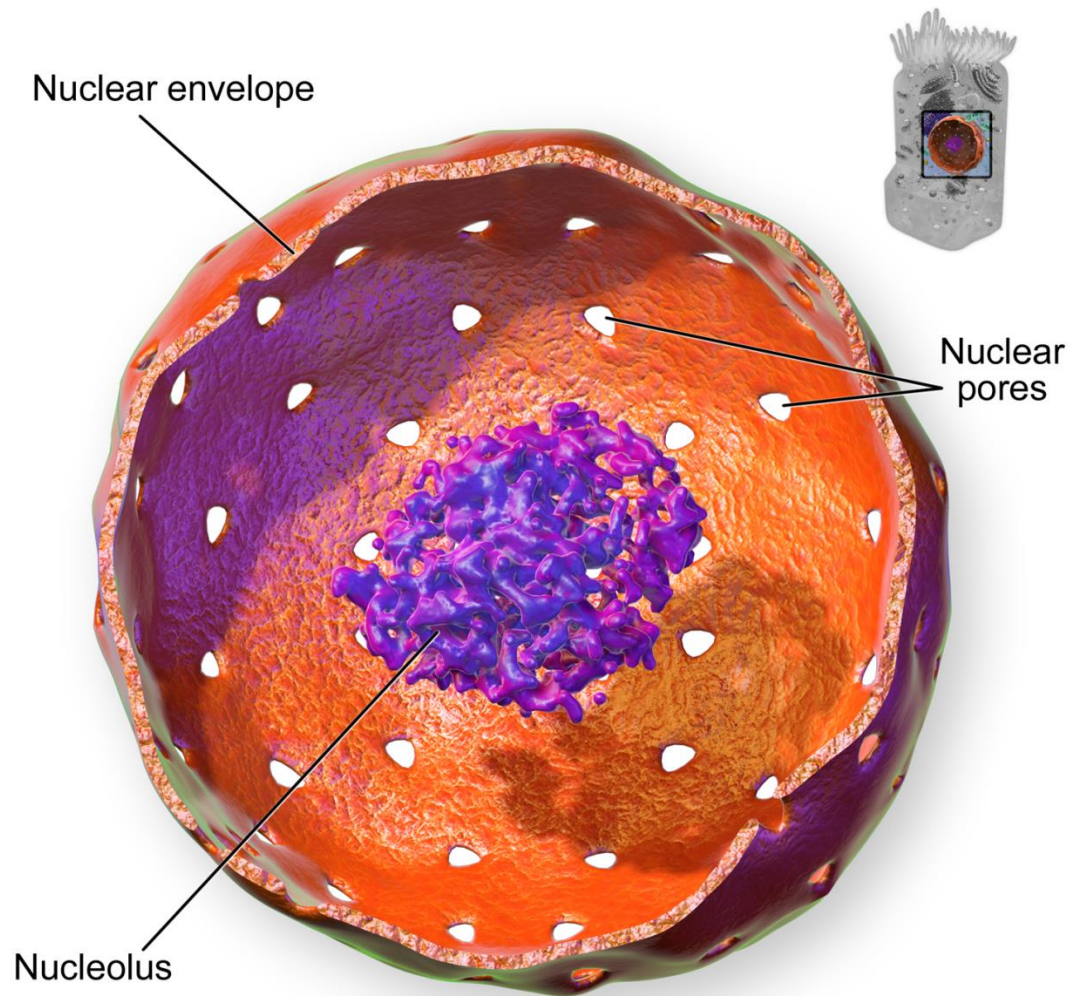
- The nucleoproteins are synthesized in the cytoplasm, then entered the nucleus while the nuclear RNA represents the newly synthesized mRNA, tRNA and rRNA which has not yet passed to the cytoplasm.

❖ Nuclear envelope:

- It is made up of **two membranes**, each about 6-8 nm thick, separated by **perinuclear space** about 40-70 nm.
- Each membrane is a **unit-membrane**.
- Its thickness by the LM is due to condensed chromatin.

- Its outer surface is studded with **ribosomes** and sometimes is continuous with rER.
- The **outer** surface is surrounded by network of **intermediate filaments**, vimentin and the **inner** surface of the inner membrane is associated with a **protein** structure called the **fibrous lamina** composed of lamina **A**, **B** and **C**.
- Several thousand of **nuclear pores** are found on the nuclear envelope.





Nucleus

Animal Cell

cell membrane

fluid-filled space

organelles

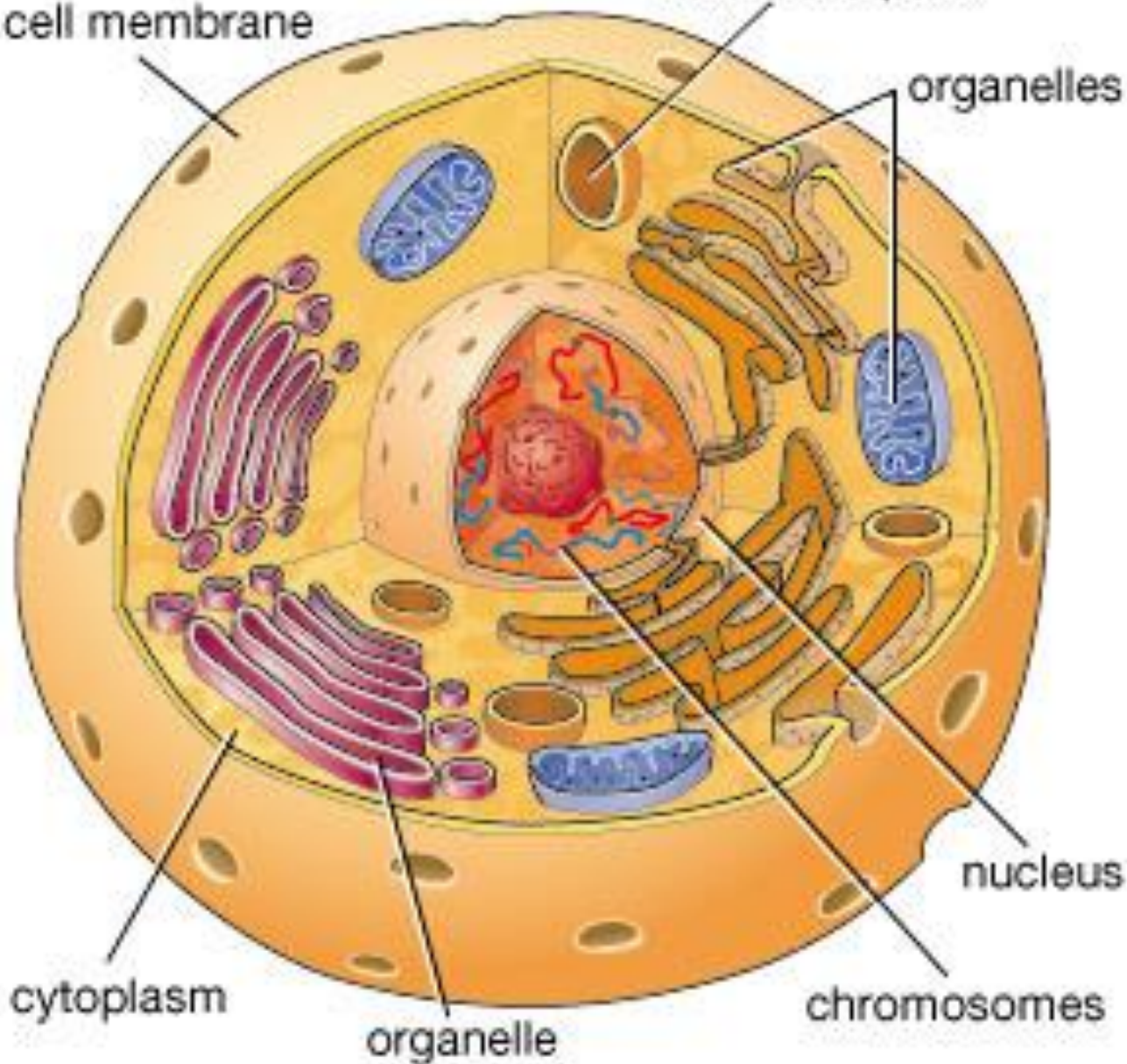
cytoplasm

organelle

nucleus

chromosomes

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❑ Nuclear pore complex

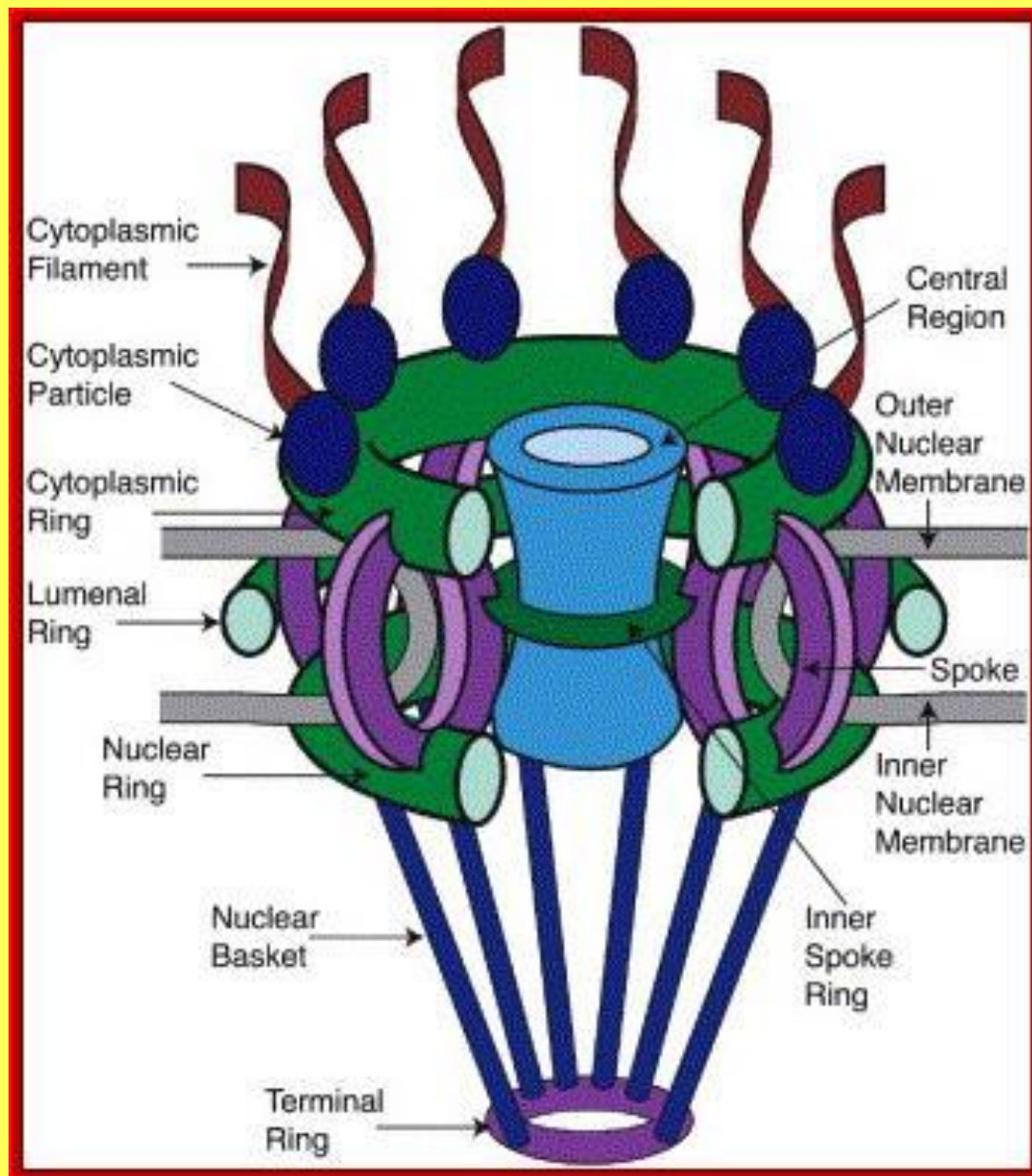
- The nuclear pore is circular and about 30-100 nm in diameter.
- At the periphery of each pore the inner and outer membranes are fused.
- The nuclear pore is surrounded by nonmembranous glycoproteins embedded in its rim forming with the pore the nuclear pore complex.
- The nuclear pore complex is about 80-100nm in diameter and composed of **three ring-like arrays of proteins**, the **cytoplasmic ring**, the **middle ring** and the **nucleoplasm ring**.

❑ Nuclear pore complex

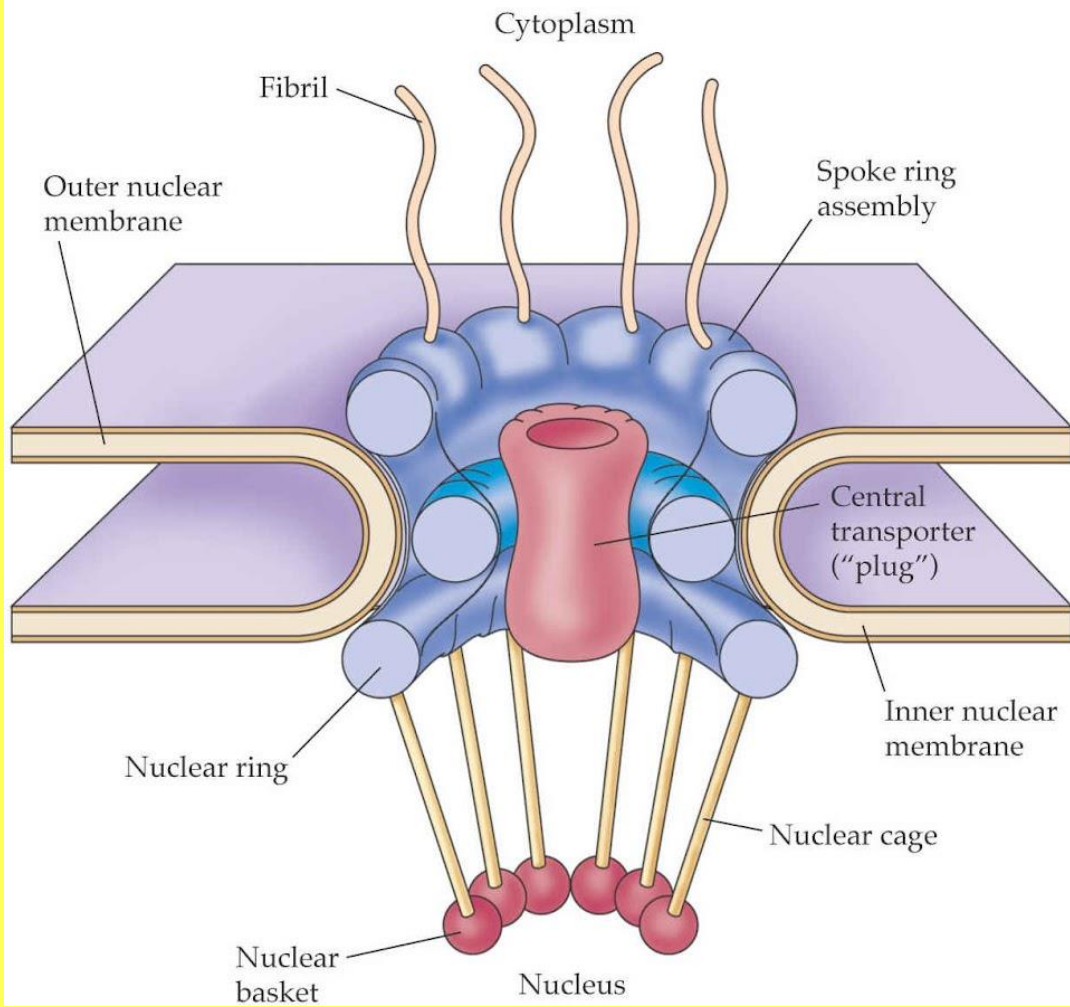
- In addition to **cytoplasmic fibers**, **transporter** and a **nuclear basket**.
- The **cytoplasmic** ring is composed of **8 subunits**, each unit possesses a filamentous fiber.
- The **middle** ring is composed of **8 spoke-like** transmembrane protein.
- The center of the **middle** ring is occupied by the transporter (**gate channel**).
- The **nucleoplasm** ring is similar to the **cytoplasmic** ring. A filamentous **nuclear basket** is suspended from this ring into the nucleoplasm.

❑ Nuclear pore complex

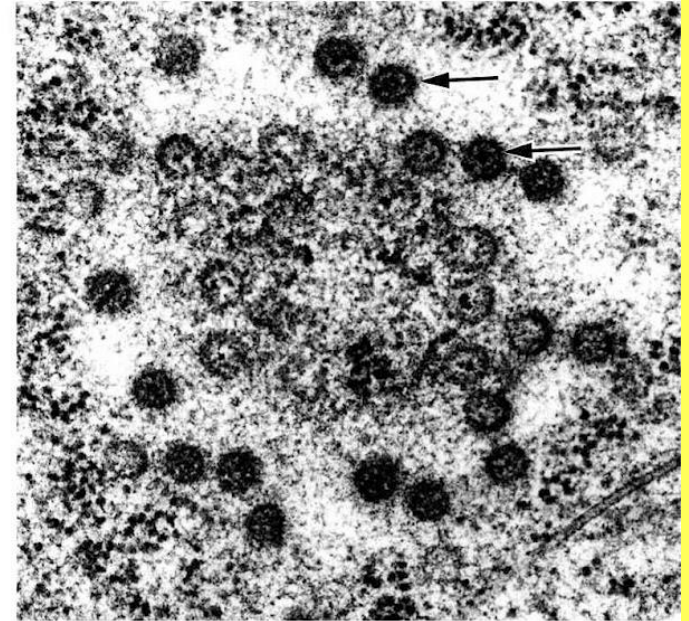
- Several **9 to 11** nm wide **channels** are available for simple diffusion however particles **larger** than 11 nm are selectively transported via a **receptor mediated transport process**.



(A)



(B)



❖ Nuclear matrix:

- It is the component that fills the space between the **chromatin** and the **nucleolus**.
- It is composed mainly of **proteins**, metabolites and ions in addition to **nucleoskeleton fibrillar framework**.
- The main component is a uniform layer of tight intermeshed **fibrillar material** that adheres to the **internal** surface of the nuclear envelope known as **fibrous lamina**.

❖ Nuclear matrix:

- The second is a **fibrogranular network** that extends from the fibrous lamina to the interior of the nucleus.
- The third is a similar framework supporting the nucleolus.
- Functions:
 - 1- Maintain the shape of the nucleus.
 - 2- It involved in gene transcription (copy of genetic information from DNA into RNA).
 - 3- It provides replication sites for DNA duplication.

❖ Chromatin

- It is a complex basophilic material of which **chromosomes** are composed.
- It is composed mainly of DNA, DNA-associated proteins (histones and non-histones) and certain amount of RNA transcribes from DNA.
- Types of chromatin:
 - 1- **Heterochromatin** (condensed) which is genetically inactive for transcription (replication).
 - It is predominant in metabolic inactive cells e.g. fibrocyte.
 - When the nucleus has large amount of heterochromatin, it is called **closed-face nucleus**.

❖ Chromatin

- Heterochromatin might be:

A- Constitutive: it is inactive all the time such as centromere.

B- Facultative: might be condensed (inactive) or decondensed (active) as X chromosome.

2- Euchromatin

- It is dispersed, **extended** or diffuse, lightly stained and **highly active** for transcription and replication.

- When a nucleus has large amount it is called vesicular or **open-face**.

- In females there is an inactive X chromosome which called **Barr body or sex chromatin**.

❖ Nucleolus

- It is a spherical basophilic structure, up to 1 μm in diameter and rich in rRNA and protein.
- It consists of:

1- Pars amorpha:

- It is the pale-staining nucleolar regions containing the nucleolar organizer DNA (5 pairs of chromosomes which carries the code for rRNA synthesis).

2- Pars nucleolonema: (nema = thread)

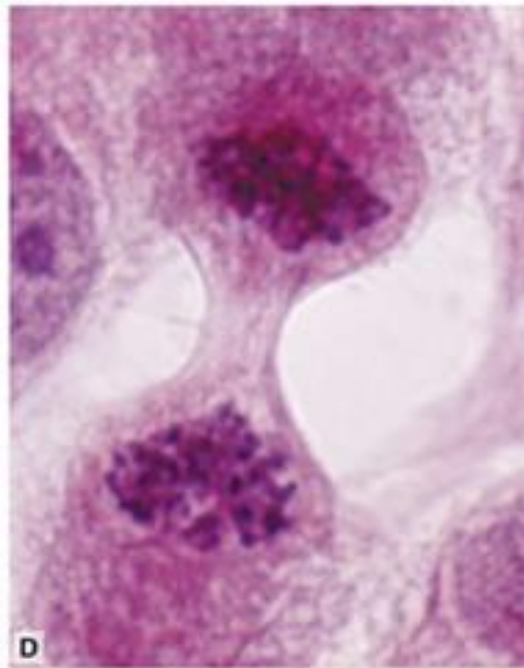
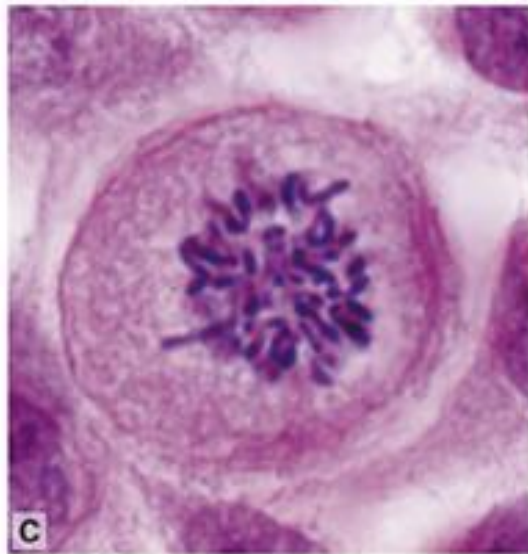
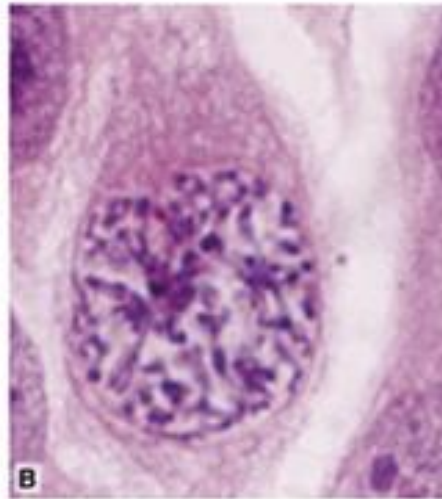
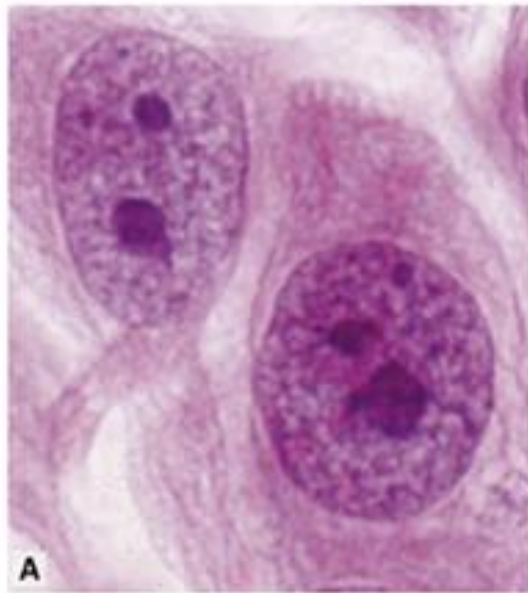
- It is a network contains two rRNA rich components:

A- The pars fibrosa, consists of fine filamentous material represent newly transcribed rRNA.

❖ Nucleolus

B- The pars granulosa, contains dense granules represent **maturing ribosomal subunits.**

- The nucleolus is the site of ribosomal RNA synthesis and ribosomal assembly.**
- The ribosomal proteins are imported from the cytoplasm and conjugated with the rRNA to form ribosomal subunits which then pass into the cytoplasm before being fully active ribosomes.**



Cell Death

- ❖ Two forms of cell death are known, **necrosis** and **apoptosis**.
- ❖ Necrosis is a pathological process that results from anoxia, mechanical injury or exposure to toxin.
- ❖ Necrotic cells and their organelles swell and finally burst releasing their content into the extracellular space. The debris of necrotic cells is taken up by macrophages.

Cell Death

- ❖ Apoptosis is a **programmed cell death**; it is not caused by injury
- ❖ It is common in cells with **short life span**.
- ❖ Unlike necrosis, apoptotic cells do not swell, but they **decrease** in size.
- ❖ The nuclei, show signs of **pyknosis** (become small and darkly-stained).
- ❖ Then the nuclei and their chromatin are broken into short pieces by endonucleases (**karyorrhexis**).
- ❖ Then the nucleus disappears (**karyolysis**), and the cytoplasm breaks into large vesicles which detach from the cell surface.
- ❖ The cell remnants are finally phagocytosed by macrophages.

Cell Death

❖ Importance of apoptosis:

- It is important in **protecting** the cells and regulating their **number**.
- Apoptosis is important in **embryo** morphogenesis.
- Most cells are able to activate their program of apoptosis (present in their chromosomes) when major changes of DNA occur just **before tumor** cells start mitosis.
- Most **T lymphocytes** are destroyed inside thymus gland before releasing into circulation to **avoid their serious damaging effect on other cells**.

